

DEPARTMENT OF LINGUISTICS & PHILOLOGY

ENGLISH PHONETICS AND PHONOLOGY

A Comprehensive Theoretical & Practical Coursebook

Academic Research & Training Initiative

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PREFACE: THE ARCHITECTURE OF SPOKEN LANGUAGE

The study of speech production has historically been divided into two separate yet deeply symbiotic academic fields: phonetics and phonology. Phonetics serves as an empirical, data-driven investigation into the physical properties of human language sounds. It observes the raw acoustic waveforms generated by speakers, the explicit physiological motions of the vocal tract, and the mechanical processes of audio reception by a listener.

Phonology, by contrast, operates in the cognitive domain. It systematically analyzes how a particular language selects a subset of these physical sounds and organizes them into a structured, rule-governed system to convey meaningful differences. This coursebook is written to address a major pedagogical gap: the distance between abstract academic theory and practical, real-world pronunciation. By integrating rigorous anatomical frameworks with structured dataset tables and transcription guides, it provides a functional map of Received Pronunciation (RP) and General American (GA) sound systems.

MODULE I: ARTICULATORY PHONETICS & THE VOCAL APPARATUS

Chapter 1: The Respiratory and Phonatory Systems

Every speech sound produced in standard English communication requires a moving column of air as its primary power source. The overwhelming majority of human speech relies on an **egressive pulmonic airstream**. This structural mechanism begins in the lungs, where the contraction of the intercostal muscles and the diaphragm forces air outward through the trachea, creating a sustainable source of aerodynamic pressure.

As this column of air journeys upward, it reaches the larynx, an anatomical structure composed of cartilages and muscles located within the neck. Inside the larynx sit the **vocal folds** (vocal cords), two parallel shelves of muscular tissue capable of varying tension and position. The space between these folds is known as the **glottis**. The state of the glottis during airflow determines the fundamental phonetic distinction of *voicing*.

States of the Glottis

1. **Voiceless State:** The vocal folds are held widely apart, allowing the air column to pass completely unobstructed into the upper cavities. Because no friction or vibration occurs at the glottis, the sound is perceived as quiet or whisper-like, such as when pronouncing the initial segment of "sit".
2. **Voiced State:** The vocal folds are brought closely together. As the egressive air is forced through this narrow constriction, it sets off a rapid, cyclical opening-and-closing vibration known as phonation. This structural resonance can be felt by touching one's throat during the pronunciation of "zoo".

Chapter 2: The Supra-Glottal Articulators

Once the air column exits the larynx, it enters the supra-glottal vocal tract, which functions as a physical acoustic resonator. This tract is divided into three primary cavities: the pharyngeal cavity, the

oral cavity, and the nasal cavity. The shape of these resonators is continuously modified by the positions of the active and passive articulators to produce specific, identifiable sound segments.

Active Articulators: These are the flexible, muscular structures of the vocal tract that move toward an anatomical target during speech production. The primary active articulator is the tongue, which linguists split into functional zones: the tip (apex), the blade (lamina), the front (dorsum), and the back (radix). The lower lip and the lower jaw (mandible) also serve as critical active articulators.

Passive Articulators: These are fixed, unmoving skeletal sites toward which the active articulators travel to narrow or close the airflow channel. Key passive structures include the upper lip, the upper teeth, the alveolar ridge (the firm, corrugated bony structure right behind the front teeth), the hard palate (the smooth bony roof of the mouth), and the soft palate or velum.

ANATOMICAL OBSERVATION DRILL 1.1

Slowly pronounce the sequence **/t/** , **/s/** , **/ʃ/** , **/k/** . Focus entirely on tracking the tactile feedback of your tongue. Identify exactly where the tongue tip or blade makes physical contact with the roof of your mouth for each individual sound segment.

MODULE II: THE CONSONANTAL MATRIX OF ENGLISH

Chapter 3: The Mechanics of Plosion (Stop Consonants)

Plosives (or oral stops) are speech sounds characterized by a complete, air-tight structural occlusion formed at some point along the vocal tract. This physical block prevents the egressive pulmonic air from escaping through either the mouth or the nose. This category of consonant production involves three distinct structural phases:

1. **The Closing Phase:** The active articulator moves toward the passive target to form a complete, secure seal.
2. **The Compression Phase:** The air column continues to push outward from the lungs, gathering behind the physical obstruction and causing an immediate spike in localized air pressure.
3. **The Release Phase:** The articulators part rapidly, allowing the compressed air to burst outward with a distinctive acoustic sound wave known as plosion.

Phonetic Symbol	Articulatory Nomenclature	Physical Mechanism of Action	Exhaustive Word Examples
/p/	Voiceless Bilabial Plosive	The upper and lower lips form a tight seal; the velum is raised to block the nasal cavity; the glottis remains open.	pie /paɪ/ , spot /spɒt/ , lip /lɪp/
/b/	Voiced Bilabial Plosive	Identical lip and velum posture as /p/, but accompanied by simultaneous vocal fold vibration.	buy /baɪ/ , cabin /ˈkæbɪn/ , rub /rʌb/
/t/	Voiceless Alveolar Plosive	The tongue tip or blade forms a complete airtight seal against the alveolar ridge; the velum is raised.	tea /ti:/ , state /steɪt/ , bat /bæt/
/d/	Voiced Alveolar Plosive	Identical lingual contact against the alveolar ridge as /t/, with active vocal fold resonance.	die /daɪ/ , ladder /ˈlædə/ , bed /bed/
/k/	Voiceless Velar Plosive	The back of the tongue (dorsum) rises to form an airtight seal against the soft palate (velum).	cat /kæt/ , market /ˈmɑːkɪt/ , back /bæk/
/g/	Voiced Velar Plosive	Identical dorsum-to-velum seal as /k/, with active vocal fold vibration.	go /ɡoʊ/ , eager /ˈiːɡəl/ , bag /bæg/

Chapter 4: Fricatives and Affricates

Fricatives differ fundamentally from plosives because they do not completely block the air column. Instead, the active articulator leaves a tiny, narrow gap against the passive target. As the egressive air is forced through this tight constriction, it speeds up significantly, creating high-velocity turbulence that results in a continuous hissing or buzzing sound wave known as friction noise.

Affricates represent complex consonants that combine elements of both plosives and fricatives within a single syllable pulse. They begin with a complete articulatory stop closure, but instead of releasing with a sudden explosion, the active articulator moves away slowly, releasing the trapped air through a narrow channel to generate localized friction noise.

Symbol	Classification	Articulatory Mechanics	Target Examples
/f/	Voiceless Labiodental Fricative	The inner margin of the lower lip is held close to the upper incisors, forcing air through.	fan /fæn/ , safe /seɪf/
/v/	Voiced Labiodental Fricative	Identical lip-to-teeth positioning as /f/, accompanied by vocal fold vibration.	van /væn/ , cave /keɪv/
/θ/	Voiceless Dental Fricative	The tip of the tongue is placed behind or between the upper and lower teeth, creating a thin gap.	thin /θɪn/ , path /pɑːθ/
/ð/	Voiced Dental Fricative	Identical interdental tongue placement as /θ/, with simultaneous glottal resonance.	this /ðɪs/ , mother /ˈmʌðə/
/s/	Voiceless Alveolar Fricative	Tongue forms a narrow central groove against the alveolar ridge, causing intense high-frequency friction.	sip /sɪp/ , face /feɪs/
/z/	Voiced Alveolar Fricative	Identical grooved tongue shape against alveolar ridge as /s/, with active vocal fold action.	zip /zɪp/ , rose /roʊz/
/ʃ/	Voiceless Post- Alveolar Fricative	The blade of the tongue forms a wide constriction slightly further back than /s/, often with rounded lips.	she /ʃiː/ , rush /rʌʃ/
/tʃ/	Voiceless Palato- Alveolar Affricate	Begins as a complete stop at the alveolar ridge, releasing slowly into a post-alveolar fricative hiss.	chair /tʃeə/ , batch /bætʃ/
/dʒ/	Voiced Palato- Alveolar Affricate	Identical stop-to-friction movement sequence as /tʃ/, with continuous vocal fold vibration.	jam /dʒæm/ , edge /ɛdʒ/

MODULE III: VOWEL SPACE GEOMETRY & LIP FORMATIONS

Chapter 5: Pure Monophthongs

Unlike consonants, which rely on clear structural contact points or narrow channels within the mouth, vowels are produced through an entirely open vocal tract. Consequently, they cannot be categorized by places or manners of constriction. Instead, the phonetic description of vowels depends on mapping a continuous three-dimensional geometric space known as the **vowel quadrilateral**. This space is defined by three main parameters:

1. **Tongue Height:** The vertical distance between the highest point of the tongue and the roof of the mouth, split into close (high), mid-close, mid-open, and open (low) positions.
2. **Tongue Backness:** The horizontal position of the highest point of the tongue body, ranging from front, to central, to back.
3. **Lip Position:** The shape of the lips during phonation, categorized as rounded (pursed forward), unrounded/spread, or neutral.

Symbol	Height / Backness Parameters	Acoustic & Mechanical Profile	Reference Lexical Set
/i:/	Close Front Unrounded (Long)	The front of the tongue rises to the highest possible position without causing friction noise; lips are spread wide.	fleece, see, beat / bi:t/
/ɪ/	Mid-Close Front-Central Lax (Short)	Slightly lower and more central than /i:/, requiring less muscular tension.	kit, sit, dim / dɪm/
/e/	Mid-Open Front Unrounded (Short)	The tongue body occupies a position halfway between close and open configurations.	dress, bet, head / hed/
/æ/	Open Front Unrounded (Short)	The jaw is lowered significantly; the tongue tip rests behind the lower incisors.	trap, hand, cat / kæt/
/ɑ:/	Open Back Unrounded (Long)	The jaw is completely lowered; the body of the tongue is flat and drawn back.	start, calm, father / 'fɑ:ðə/
/ɔ:/	Mid-Open Back Rounded (Long)	The tongue is pulled back with a mid-low elevation; lips are clearly rounded.	thought, saw, law / 'lɔ:/
/ʊ/	Mid-Close Back Rounded (Short)	A lax back vowel where the tongue is slightly raised toward the soft palate; lips are loosely rounded.	foot, put, good / gʊd/
/u:/	Close Back Rounded (Long)	The back of the tongue rises high toward the velum; lips are tightly rounded.	goose, blue, boot / 'bu:t/
/ʌ/	Mid-Open Central-Back Unrounded	The tongue rests in a low-central position with a completely relaxed jaw and neutral lip profile.	strut, mud, cup / kʌp/
/ə/	Mid Neutral Central (The Schwa)	The ultimate unaccented placeholder sound in English. It requires zero muscular effort and occurs only in unstressed positions.	about / 'ə'baʊt/ , banana / bə 'nɑ:nə/

MODULE IV: SUPRASEGMENTALS & DYNAMIC CONNECTED SPEECH

Chapter 6: Syllabic Structures and Stress Laws

Human speech does not consist of individual, isolated sound segments spoken one after another. Instead, sounds are combined into rhythmic pulses known as **syllables**. The structural anatomy of an English syllable follows a strict hierarchical organization:

The core of every syllable is the **Nucleus**, which is typically a vowel sound or a syllabic consonant (like the final sound in "button"). Any consonants that appear before the nucleus form the **Onset**, while any consonants that follow the nucleus form the **Coda**. Together, the nucleus and the coda make up the **Rime** of the syllable.

In words that contain more than one syllable, English applies strict acoustic hierarchies. **Primary Word Stress** (marked with the symbol **/** placed before the target syllable) means that a specific syllable is produced with greater respiratory energy, higher muscular tension, increased duration, and a distinct pitch movement.

Chapter 7: Boundary Modulations (Assimilation, Elision & Linking)

When words are combined into fluent, rapid speech, the articulators do not have enough time to reach the ideal textbook positions for every single sound segment. To save energy, adjacent sounds adjust to one another across word boundaries.

Assimilation: This occurs when a speech sound changes its physical properties to match or resemble a neighboring sound segment. A common example is *regressive place assimilation*, where an alveolar sound like **/n/** or **/t/** shifts its point of contact to match a following bilabial or velar sound (e.g., "ten boys" often shifts naturally in fluent speech to sound like "tem boys").

Elision: This refers to the complete disappearance of a speech sound that would be present if the word were spoken in isolation. This regularly happens to the sounds **/t/** and **/d/** when they appear in complex consonant clusters at the end of a word (e.g., "last night" sounds like "las night").

Linking (Liaison): To avoid abrupt pauses between words, fluent English inserts specific transitional sounds across word boundaries. The most common form is the *linking 'r'*, where a word that ends with a silent 'r' in isolation pronounces that 'r' when the next word begins with a vowel sound (e.g., "four" **/fɔ:/** becomes "four apples" **/fɔ:r 'æpəlz/**).

PHONETIC TRANSCRIPTION CHALLENGE

Transcribe the following full sentence into connected speech IPA, incorporating appropriate weak forms, elisions, and linkings: *"Tell the teacher that the next door neighbor is buying four apples."*